PROJECT WORK ALLOCATION PLAN

FitBuddy

AI Fitness Plan Generator using Gemini Models

TEAM ID	TOTAL MEMBERS	CORE ENGINE
SWTID-2026-6912	5 Team Members	Google Gemini API

INDIVIDUAL ROLES & TASK RESPONSIBILITIES

1. Pavithra M

Team Lead & AI Integration Lead

- Overall project leadership, coordination, and timeline management.
- Design and integration of the core Google Gemini API architecture.
- Prompt engineering for personalized workout & dietary recommendation engines.
- Handling API key security, request throttling, and model response parsing.

2. Manisha S

Frontend UI/UX Developer

- Designing intuitive user interfaces for profile creation and survey inputs.
- Building responsive dashboard components to display daily AI fitness routines.
- Implementing interactive input forms (Age, Weight, Fitness Goals, Dietary Preferences).
- Ensuring seamless user experience and accessibility across devices.

3. Monica G

Backend & Database Engineer

- Architecting backend RESTful APIs to connect UI with Gemini LLM services.
- Database schema design for storing user profiles, history, and generated plans.
- User authentication and session management implementation.
- Developing server-side logging and analytics services.

4. Monisha Anandan

Fitness & Nutrition Logic Specialist

- Formulating domain logic for calorie, BMI, BMR, and macro-nutrient calculations.
- Building rule-based validation pipelines for Gemini output verification.
- Creating safety guardrails and health disclaimers for AI fitness routines.
- Structuring predefined template options for target fitness tiers (Beginner/Intermediate/Advanced).

5. Mridula M

QA, Testing & Documentation Lead

- Conducting unit, integration, and user-acceptance testing (UAT).
- Testing AI model response reliability, edge cases, and safety compliance.
- Preparing comprehensive software documentation, project report, and user guide.
- Creating final presentation deck and video demonstration scripts.